

Software Architecture Assignment 3: Architectural Modeling and Design

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1. Introduction and System Overview

Selected System

The selected system is a **Geolocation-Based Service Worker Marketplace Platform**.

System Overview

The platform connects customers who require immediate or scheduled services (such as plumbing, electrical work, cleaning, and maintenance) with nearby service workers using real-time geolocation. Customers can create service requests, discover and compare available workers, communicate within the platform, track job progress, and complete payments. Service workers manage profiles, respond to requests, perform services, and track earnings. Administrators supervise user verification, dispute resolution, and overall platform operation.

The system acts as an intermediary marketplace, ensuring efficient matching between service demand and supply while maintaining trust, transparency, and reliability for all stakeholders.

2. Domain Model

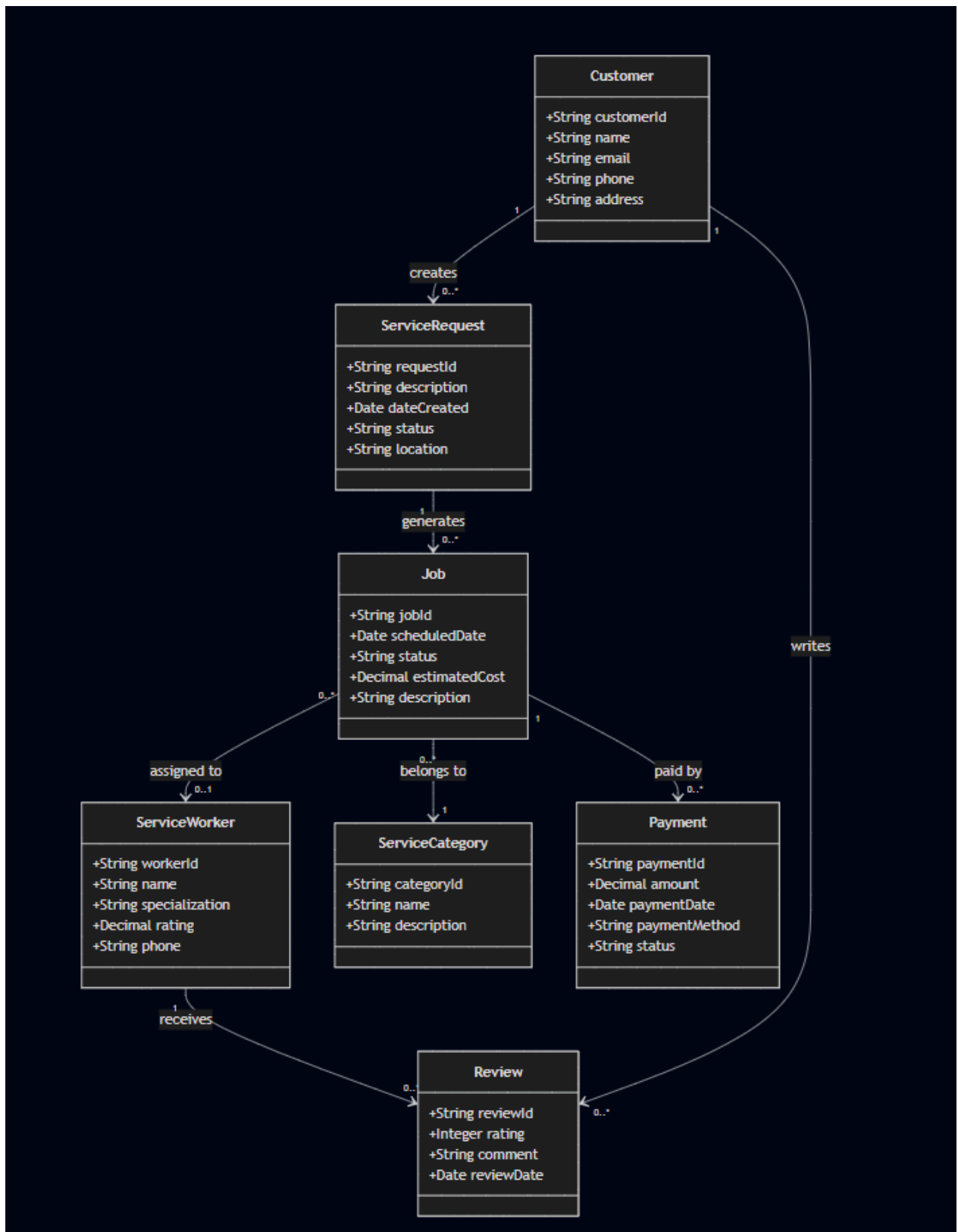
2.1 Purpose

The domain model represents the core business concepts of the service marketplace using non-technical language understandable to all stakeholders.

2.2 Key Domain Entities

- **Customer:** A person requesting a service.
- **Service Worker:** A professional offering services.
- **Service Request:** A description of work needed by a customer.
- **Service Category:** Classification of services (e.g., plumbing, cleaning).
- **Job:** An accepted service request being executed.
- **Payment:** Compensation for a completed job.
- **Review:** Feedback provided after job completion.

2.3 Domain Model Diagram



2.4 Business Rules

1. A job can only be created if a service worker accepts a service request.
2. Payment is processed only after a job is marked as completed.
3. A customer can leave a review only for completed jobs.

2.5 Explanation

These entities were selected because they represent the fundamental business concepts required to operate a service marketplace. The domain model clarifies system requirements by defining responsibilities, constraints, and interactions in business terms, helping align stakeholder expectations.

3. Informational Model

3.1 Purpose

The informational model describes how information is structured and managed by the system, focusing on data rather than business meaning.

3.2 Informational Model Diagram (ASCII)

3.3 Data Constraints

- Rating values must be between 1 and 5.
- Payment amount must be greater than zero.
- Job status follows a strict lifecycle.

3.4 Comparison with Domain Model

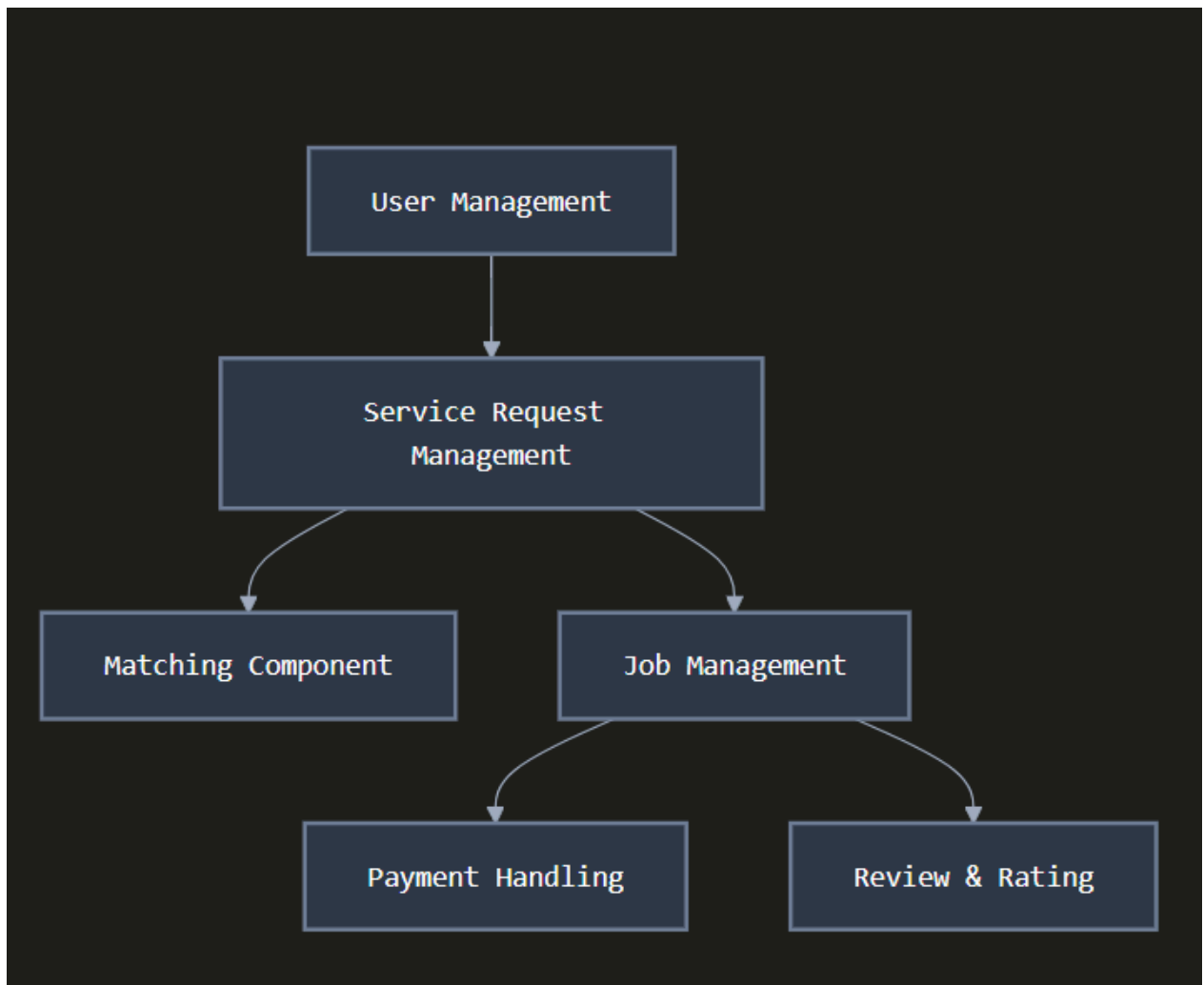
The domain model explains *what* the business concepts are, while the informational model explains *how* information is structured. Both are required to avoid ambiguity between business meaning and data representation.

4. Design Model

4.1 Purpose

The design model presents a high-level view of the system's internal structure and responsibilities.

4.2 Design Model Diagram

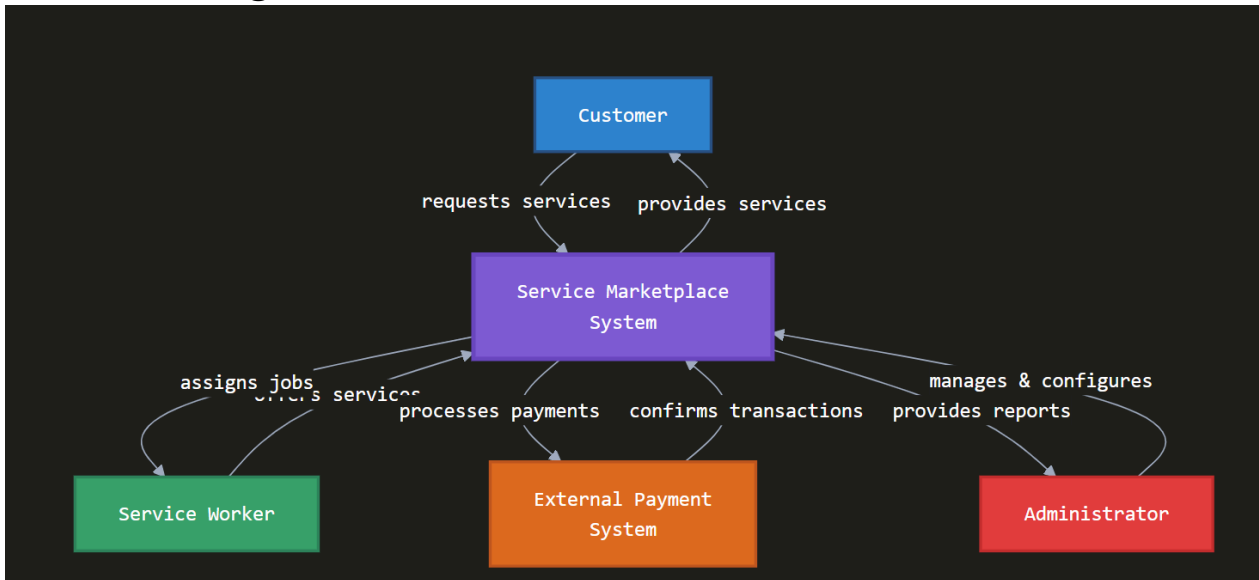


4.3 Explanation

Each component has a clearly defined responsibility. Interactions reflect the logical flow of service request creation, matching, execution, payment, and feedback, without exposing implementation details.

5. Context Diagram

5.1 Context Diagram



5.2 Validation

This diagram qualifies as a context diagram because it represents the system as a single unit and shows only external actors and systems. Internal components and technologies are intentionally excluded.

6. Architectural Design Decisions

Decision 1: Centralized Matching Logic

- **Context:** Efficiently pairing customers with nearby workers is critical.
- **Problem:** Where should matching responsibility reside?
- **Options:** Distributed matching vs centralized matching.
- **Decision:** Centralized matching component.
- **Consequences:** Simplifies control and consistency but may require careful scaling.

Decision 2: Separation of Job and Payment Responsibilities

- **Context:** Jobs and payments have different lifecycles and rules.
- **Problem:** Should job execution and payment be handled together?
- **Options:** Single combined component vs separate components.
- **Decision:** Separate Job Management and Payment Handling components.
- **Consequences:** Improves clarity and maintainability at the cost of additional coordination.

7. Model Consistency Analysis

All models describe the same core concepts at different abstraction levels. Domain entities map directly to informational data entities and are realized by design components. Inconsistencies could lead to misunderstandings, redundant data, or incorrect responsibilities. Maintaining alignment ensures architectural clarity.

8. Assumptions and Open Questions

Assumptions

- Users are willing to share location data.
- Payments are handled by trusted external providers.

Open Questions

- How should disputes between customers and workers be resolved?
- Should different service categories follow different job lifecycles?

Conclusion

This assignment demonstrates the use of multiple architectural models to describe a single system, highlighting how different perspectives address different concerns while remaining consistent and complementary.